KAZAKH-BRITISH TECHNICAL UNIVERSITY

JSC «Kazakh British Technical University»
School of IT and Engineering

APPROVED BY
Dean of SITE
Azamat Imanbayev

«05» 01 2026

SYLLABUS

Discipline: CSCI 2109 IT Infrastructure and Computer Networks
Term: Spring 2026
Instructors' full names: Mels Begenov, Temirlan Zhaxalykov, Yerlan Sharipov, Azamat Imanbayev, Zhanel Zhakhayeva, Viktoriya Kalashnikova

Personal Information about the Instructor	Time and place of classes		Contact information
	Lessons	Office Hours	e-mail
Mels Begenov, Senior Lecturer	According to the schedule		m.begenov@kbtu.kz
Yerlan Sharipov Senior Lecturer	According to the schedule		y.sharipov@kbtu.kz
Temirlan Zhaxalykov Senior Lecturer	According to the schedule		t.zhaxalykov@kbtu.kz
Azamat Imanbayev, Assistant Professor	According to the schedule		a.imanbaev@kbtu.kz
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Viktoriya Kalashnikova Tutor	According to the schedule		v.kalashnikova@kbtu.kz

COURSE DURATION: 6 ECTS, 15 weeks, 60 class hours

GENERAL COURSE AIMS:

The general aim of this course is to equip students with foundational knowledge and practical skills in computer networking, focusing on the architecture, components, and operational principles of the Internet and other networks. It aims to enable students to design and implement basic network infrastructures, configure essential network devices, and apply IP addressing schemes to support effective communication and connectivity.

COURSE DESCRIPTION

This course introduces the architecture, structure, functions, components, and models of the Internet and other computer networks. The principles and structure of IP addressing and the fundamentals of Ethernet concepts, media, and operations are introduced to provide a foundation for the curriculum. By the end of the course, students will be able to build simple LANs, perform basic configurations for routers and switches, and implement IP addressing schemes.

COURSE OBJECTIVES

Students who complete "Computer Networks" will be able to perform the following functions:

- Understand and describe the devices and services used to support communications in data networks and the Internet
- Understand and describe the role of protocol layers in data networks
- Understand and describe the importance of addressing and naming schemes at various layers of data networks in IPv4 and IPv6 environments
- Design, calculate, and apply subnet masks and addresses to fulfill given requirements in IPv4 and IPv6 networks
- Explain fundamental Ethernet concepts, such as media, services, and operations
- Build a simple Ethernet network using routers and switches
- Use command-line interface (CLI) commands to perform basic router and switch configurations

Utilize common network utilities to verify small network operations and analyze data traffic

COURSE OUTCOMES

By the end of this course, students will be able to:

1. Understand and describe the architecture, components, and models of computer networks, including the Internet.
2. Explain the principles and structure of IP addressing and subnetting.
3. Demonstrate knowledge of Ethernet concepts, media, and operations.
4. Design and build simple LANs (Local Area Networks).
5. Configure basic settings on routers and switches to establish network connectivity.
6. Implement and troubleshoot IP addressing schemes in network environments.
7. Analyze and solve fundamental networking problems using appropriate tools and techniques.
8. Apply foundational networking skills to real-world scenarios, preparing for further advanced studies in networking or related fields.
9. Explain the role and operation of switches, routers, and wireless devices in small to medium-sized networks.
10. Configure and troubleshoot VLANs, trunking, and inter-VLAN routing.
11. Implement static and dynamic routing protocols to enable network connectivity.
12. Describe and configure basic wireless network concepts, standards, and security settings.
13. Apply redundancy and high-availability concepts to improve network reliability.
14. Configure and troubleshoot DHCP, NAT, and basic network services.
15. Use network diagnostic tools to identify and resolve common network issues.

16. Apply switching, routing, and wireless skills to design and support functional enterprise LANs.
17. Analyze the architecture and design principles of enterprise-scale networks.
18. Configure and troubleshoot advanced routing technologies, including OSPF and WAN solutions.
19. Implement network security concepts such as access control lists (ACLs), firewall basics, and VPN technologies.
20. Explain and apply Quality of Service (QoS) concepts to manage network traffic effectively.
21. Monitor and troubleshoot enterprise networks using appropriate tools and methodologies.
22. Describe network virtualization, automation, and programmability concepts.
23. Use basic automation techniques and controllers to manage and configure network devices.
24. Apply enterprise networking, security, and automation skills to real-world scenarios and prepare for professional networking roles.

COURSE PREREQUISITE: Computer architecture
COURSE POST REQUISITES: Advanced Networking, Network Security

Literature

Required

1. Cisco Networking Academy "CCNA Introduction to Networks" curriculum
2. CCNA Routing, Switching and Wireless Essentials
3. CCNA Enterprise Networking, Security and Automation
4. CCST Networking Essentials
5. Олифер Виктор, Олифер Наталья. Компьютерные сети. Принципы, технологии, протоколы: Юбилейное издание. — СПб.: Питер, 2021. — 1008 с.:
6. Doug Lowe. Networking All-in-One Desk Reference For Dummies: 8th Edition. Wiley, 2021
7. James Kurose, Keith Ross. Computer Networking: A Top-Down Approach, 7th Edition. Pearson, 2017
8. Mike Meyers. CompTIA Network+ Certification All-in-One Exam Guide. McGraw-Hill Education, 2018
9. Andrew S. Tanenbaum, "Computer Networks", 4rd Edition, Prentice Hall PTR, 2003.

COURSE ASSESSMENT CRITERIA

Assessment occurs continuously throughout the course. The evaluation will be based on the levels of (maximums in %):

Type of activity	Final scores
Attendance /participation	**0**
Practice	**30**
Midterm/Endterm	**30**
Final exam	**40**
Total	**100**

TASKS

for practice

Week	Practice	Cost (in points)
2	Practice 1	3
3	Practice 2	3

4	Practice 3	3
5	Practice 4	3
7	Practice 5	3
9	Practice 6	3
10	Practice 7	3
11	Practice 8	3
12	Practice 9	3
13	Practice 10	3
	Total	**30**

COURSE CALENDAR

Week	Class work		
	Topic	**Lectures**	**Practice**
1	**Lecture 1: Introduction to networks** Networks Affect our Lives Network Representations and Topologies Common Types of Networks Internet Connections Reliable Networks Network Trends Network Security	2	1
2	**Lecture 2: Cisco: Basic IOS Configurations** Cisco IOS Access IOS Navigation The Command Structure Basic Device Configuration Save Configurations Ports and Addresses Configure IP Addressing **Practice 1:**	2	1
3	**Lecture 3: Physical Layer/Data link Layer** Purpose of the Physical Layer Physical Layer Characteristics Copper Cabling UTP Cabling Fiber-Optic Cabling Wireless Media Purpose of the Data Link Layer Topologies Data Link Frame **Practice 2:**	2	1

4	**Lecture 4: VLANs / Inter-Vlan routing / Etherchannel** VLANs VLAN in a Multi-Switched Environment VLAN Configuration VLAN Trunks Dynamic Trunking Protocol Inter-VLAN Routing Operation Router-on-a-Stick Inter-VLAN Routing Inter-VLAN Routing using Layer 3 Switches EtherChannel Operation Configure EtherChannel **Practice 3:**	2	1
5	**Lecture 5: Switching Concepts / STP concepts** Frame Forwarding Collision and Broadcast Domains Purpose of STP STP Operations Evolution of STP **Practice 4:**	2	1
6	**Lecture 6: Ethernet Switching / Address Resolution** Ethernet Frames Ethernet MAC Address The MAC Address Table Switch Speeds and Forwarding Methods MAC and IP ARP IPV6 Neighbor Discovery **Practice 5:**	2	1
7	Midterm	2	1
8	**Lecture 7: Network Layer/IPv4 Addressing** Network Layer Characteristics IPv4 Packet IPv6 Packet IPv4 Address Structure Types of IPv4 Addresses Network Segmentation VLSM	2	1

9	**Lecture 8: IPv6 Addressing/ICMP** IPv4 Issues IPv6 Address Representation IPv6 Address Types GUA and LLA Static Configuration Subnet an IPv6 Network ICMP Messages Ping and Traceroute Tests **<u>Practice 6:</u>**	2	1
10	**Lecture 9: Routing Concept / IP Static Routing** Path Determination Packet Forwarding Basic Router Configuration Review IP Routing Table Static and Dynamic Routing Static Routes Configure IP Static Routes Configure IP Default Static Routes Configure Floating Static Routes Configure Static Host Routes **<u>Practice 7:</u>**	2	1
11	**Lecture 10: OSPF** OSPF Features and Characteristics OSPF Packets OSPF Operation OSPF Router ID Point-to-Point OSPF Networks Multiaccess OSPF Networks Modify Single-Area OSPFv2 Default Route Prorogation **<u>Practice 8:</u>**	2	1
12	**Lecture 11: NAT** NAT Characteristics Types of NAT NAT Advantages and Disadvantages Static NAT Dynamic NAT PAT NAT64 **<u>Practice 9:</u>**	2	1

13	**Lecture 12: Transport Layer** Transportation of Data TCP/UDP Overview Port Numbers TCP Communication Process Reliability and Flow Control **<u>Practice 10:</u>**	2	1
14	**Lecture 13: Application Layer** Application, Presentation, and Session Peer-to-Peer Web and Email Protocols IP Addressing Services File Sharing Services	2	1
15	Endterm	2	1
16-17	Final Exam	30	15

Course assessment schedule

No	Assessment criteria	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
1	Attendance / participation	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*	**0%**
2	Practice		*	*	*	*		*		*	*	*	*	*			**30%**
3	Midterm/ End-term								*							*	**30%**
4	Final exam																**40%**
	Total																**100%**

Class sessions – will be a mixture of information, discussion and practical application of skills.
Practice – will reinforst the students knowledge by practical appliance of lectured materials.
In-class assessment – will prepare students for their mid-term and final assessment and identify the competence level they have achieved on a related subject matter, the aim being to diagnose potential discrepancies in students' understanding and performance in order to make specific adjustments to the course content and procedures and/or to assign additional assignments to certain individuals or the whole group.
Home assignments – will consolidate the concepts and materials taken during in-class activities, help students to expand the content through diverse background resources and/or practise certain skill areas; they will also develop the students' ability to work individually in exploring and examining related issues.
SIS (Student Independent Study) – comprises group Project to be done by students on the independent basis. Students are supposed to use knowledge and skills acquired in class to do the project. Assistance and advice will be provided by teachers during office hours.
TSIS (Teacher Supervised Student Independent Study) – student self-made project.

End-term test – a diagnostic test used to identify the students' progress, their strengths and weaknesses, intended to force student to prepare for Final Exam. It includes computer based test.

Final examination – 1) an attainment test designed to identify how successful the students have been achieving objectives.

Grading policy:

Intermediate attestations (on 8th and 15th week) join topics of all lectures, practice, laboratories, SIS, TSIS and materials for reading discussed to the time of attestation. Maximum number of points within attendance, activity, SIS, TSIS and laboratories for each attestation is 30 points.

Final exam joins and generalizes all course materials, is conducted in the complex form with quiz and problem. Final exam duration is 100 min. Maximum number of points is 40. At the end of the semester you receive overall total grade (summarized index of your work during semester) according to conventional KBTU grade scale.

ATTENTION!

1) If student missed without plausible reason more than **30% of lessons student receives «F (Fail)»** grade;
2) If for two attestations student receives 29.4 or less points, this student is not accepted to final exam and for all course he (she) receives **«F (Fail)» grade;**
3) If student receives 9.4 or less points on final exam, then independently on how many points he (she) received for two attestations, in whole he (she) receives **«F (Fail)» grade;**

In the case of missing or being late for final exam without plausible reason, independently on how many points he (she) received for two attestations, in whole he (she) receives **«F (Fail)» grade.**

4) Practical work must be uploaded to the Teams assignment in written form. If a student fails to upload the work before the specified deadline, the work will not be considered. A score of zero is given.
5) All written work must be completed according to the **standard** of KBTU. The standard and template are attached to the TEAMS platform. The **MS Teams team** has added a video lesson on preparing reports according to standards.
6) All text materials are available on the https://wsp.kbtu.kz/ and TEAMS platform.
7) If a student does not pass the **practice task** on time, then admission to the next **practice** will be only after the mandatory completion of the previous work. Points for previous work will be lost, but feedback will be given on correct completion.
8) If plagiarism is detected in the text of the work, in the program code, or if the work does not meet the standards, the work is canceled.

ACADEMIC POLICY:

1. Cheating, duplication, falsification of data, plagiarism are not permitted under any circumstances!
2. Students must participate fully in every class. While attendance is crucial, merely being in class does not constitute "participation". Participation means reading the assigned materials, coming to class prepared to ask questions and engage in discussion.
3. Students are expected to take an active role in learning (the instructor will provide the information and guidelines to do this).
4. Students must come to class on time.
5. Students are to take responsibility for making up any work missed.
6. Make up tests in case of absence will not normally be allowed.
7. Mobile phones must always be switched off in class.
8. Students should always show tolerance, consideration and mutual support towards other students.

Grade		Achievement percentage	Assessment criterion
«Excellent»	A	95-100%	This grade is given when the student: demonstrated a complete understanding of the course material; did not make any errors or inaccuracies; completed control and laboratory work in a timely and correct manner, and submitted reports on them; demonstrated original thinking; submitted control quizzes on time and without any errors; completed homework assignments; engaged in research work; independently used additional scientific literature in studying the discipline; was able to independently systematize the course material.
	A -	90-94%	
«Good»	B+	85-89%	This grade is given when the student: Has mastered the course material at no less than 75%; Did not make gross errors in responses; Timely completed control and laboratory work and submitted them without fundamental remarks; Correctly completed and timely submitted control tests and homework assignments without fundamental remarks; Utilized additional literature as indicated by the instructor; Engaged in research work, made non-fundamental errors, and fundamental errors corrected by the student themselves; Managed to systematize the course material with the help of the instructor.
	B	80-84%	
	B-	75-79%	
	C+	70-74%	
«Satisfactory»	C	65-69%	This grade is given when the student: Has mastered the course material no less than 50%; Required assistance from the instructor when completing control and laboratory work, homework assignments; Made inaccuracies and non-fundamental errors when submitting control tests; Did not demonstrate activity in research work, relied solely on the educational literature indicated by the instructor; Experienced more difficulty in systematizing the material.
	C-	60-64%	
	D+	55-59%	
	D	50-54%	

Senior Lecturer **Mels Begenov**
Senior Lecturer **Yerlan Sharipov**
Senior Lecturer **Temirlan Zhaxalykov**

Minutes # 9 of the School of Information Technology and Engineering meeting on January 5, 2026